When Demonstrations Lose to Priors

Demonstration Selection for In-Context Learning under Distribution Shift in 7B Language Models

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This thesis is submitted in partial fulfillment of
the requirements for the degree of
Bachelor of Advanced Computing (Honours)

School of Computer Science
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13 October 2026



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Abstract

Large language models can learn a task from a handful of labelled examples placed in their prompt, a behaviour called in-context learning. Which examples are shown matters, and a growing literature selects them by similarity, diversity or learned scores. Little is known about whether selection helps when the test data shift away from the examples, or when what the examples teach contradicts what the model already believes. Smaller models are known to follow such contradicting evidence only partially.

This thesis asks whether choosing the demonstrations can help a 7B model follow them over its own priors under controlled distribution shift. We build a synthetic tabular testbed in which covariate shift, spurious-feature reversal and concept shift (realised by giving features names whose everyday meaning agrees or disagrees with the rule) can each be switched on alone, with known ground truth. On Qwen2.5-7B-Instruct with eight demonstrations, we compare seven hand-designed selection strategies against a zero-shot baseline. We then train a set-transformer selector (SATA), first on structural targets and then on the model's own measured priors (SATA-MI), and look inside the model with the logit lens, linear probes and activation steering. Every confirmatory test was written down before its data were seen.

Feature names, not demonstrations, decide most of what the model predicts. Flipping the names costs 0.20 AUROC (95% CI [0.13, 0.28], 22 of 24 tasks), more than any shift of the data. No hand-designed strategy overcomes it. **A trained selector beats every strategy on a stand-in learner, but not on the instruction-tuned model**: none of the six pre-registered SATA contrasts, and none of the four SATA-MI contrasts, favours the selector after Holm correction. In an exploratory comparison, SATA does help the base model, whose priors pull less. **Inside the model, the answer forms only in its last few layers, and that is where the names take over**: its zero-shot guess is decodable from layer 2, the demonstrated rule hardly at all, and a steering direction for "follow the demonstrations" moves its outputs without making them more correct.

Within the scale we could study (7B models, eight numeric demonstrations, synthetic tasks, a small compute budget), selection alone cannot route around a strong prior. The testbed, the contrast families and the diagnosis give future work at larger scale a controlled place to test what can.

Acknowledgements

[1]

[1] **TODO** *Acknowledgements: supervisor, School of Computer Science, anyone who gave feedback, compute providers.*

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CHAPTER 1

Introduction

A large language model can be taught a new classification task without changing a single weight. Place a few labelled examples (*demonstrations*) in the prompt, follow them with an unlabelled *query*, and the model will usually answer in kind (Brown et al., 2020; Dong et al., 2024). This *in-context learning* is cheap, which is why it is used so widely, and it is fragile: which demonstrations are shown, in what order and with what label balance, changes the answer (Lu et al., 2022; Min et al., 2022; Zhao et al., 2021). A body of work therefore *selects* demonstrations, by similarity to the query (Liu et al., 2022), by diversity and coverage (Gupta et al., 2023; Levy et al., 2023), or with retrievers trained on the model's own feedback (Li et al., 2023; Rubin et al., 2022; Ye et al., 2023).

Two things that matter in practice are missing from most of that work. The first is **distribution shift**: the queries a deployed model sees are rarely drawn from the same distribution as the demonstrations. The second is **prior conflict**: a model brings its own beliefs about what a feature means, and what the demonstrations teach can disagree with them. Wei et al. (2023) found that following in-context evidence that contradicts semantic priors emerges with scale, and models of 7–8B parameters follow contradicting demonstrations only partially (Kossen et al., 2024; Krishna Kumar, 2025). Yet these are the models that most practitioners can afford to run.

1.1 The question

This thesis asks a single question:

> *Can the choice of demonstrations help a 7B language model follow in-context evidence over its own priors, under controlled distribution shift?*

It is broken into four research questions, each answered on Qwen2.5-7B-Instruct (Qwen et al., 2024) with $k = 8$ demonstrations.

RQ1:

Does in-context learning degrade under covariate shift, spurious-feature reversal and concept shift?

RQ2:

Do different selection strategies help under different kinds of shift?

RQ3:

Does the model rely on the features that actually determine the label, and why does its prior win when it conflicts with the demonstrations?

RQ4:

Does a learned selector (SATA) beat the hand-designed strategies, and do signals measured from the model itself help it?

Answering these questions on real data is not possible, because real benchmarks mix several kinds of shift and give no ground truth for which features matter (section 2.3). We therefore build a **synthetic tabular testbed** (Chapter 3). Every task has three features that determine its label, a spurious "shortcut" feature that agrees with the label 80–90% of the time, and distractors. Covariate shift, reversal of the shortcut and concept shift can each be switched on alone. Concept shift is realised by *naming*: the same task is shown with neutral names ($f_0, \ldots, f_9$), with everyday names whose meaning agrees with the rule (*aligned*), or with the antonyms of those names (*flipped*), so that the model's semantic prior points the wrong way.

1.2 Approach and the story of this thesis

The experiments form a chain, and each chapter of Part II starts from the limitation of the one before it.

(1) **Shift, names and the prior** (Chapter 4; **RQ1**–**RQ3**). We measure how each shift hurts and whether seven hand-designed strategies help. The names dominate: no strategy overcomes flipped names, and the model's "bigger numbers mean 1" habit explains much of what remains. *Limitation:* hand-designed heuristics see only the data, not what the model makes of it.

(2) **Learning to select** (Chapter 5; **RQ4**). We train a set transformer, SATA, to choose demonstrations that teach the rule rather than the shortcut, and then a variant, SATA-MI, that also reads the model's own measured prior for every candidate and the query. Both beat every strategy on stand-in learners and tie on Qwen-Instruct; SATA does help the base model. *Limitation:* behaviour alone cannot tell us *why* no choice of evidence works.

(3) **Inside the model** (Chapter 6; **RQ3**). The logit lens, linear probes and steering show that the answer forms in the last few layers, that the names decide it there, and that the demonstrated rule barely reaches the residual stream. *Limitation:* our scale: one 7B model, eight numeric demonstrations, synthetic tasks.

1.3 Contributions

(1) **A controlled testbed for in-context learning under shift and prior conflict**, with each shift isolated, ground-truth feature roles, a naming factor matched on measured prior strength, and a statistical protocol (within-cell AUROC, a hierarchical bootstrap over tasks, Holm-corrected contrast families frozen before the data) (Chapter 3).

(2) **A measured account of how a 7B model trades demonstrations against priors**: names outweigh every data shift, and the model's numeric prior explains most of the cost of reversing the shortcut (Chapter 4).

(3) **SATA and SATA-MI**, demonstration selectors trained on structural and model-informed targets, with pre-registered evaluations whose null results on the instruction-tuned model are reported as such (Chapter 5). As far as our literature search reached, this is the first selector trained on a model's measured semantic priors and tested under controlled shift.

(4) **A mechanistic diagnosis** of why selection fails at this scale, and a negative result for single-direction steering (Chapter 6).

1.4 Summary of findings

Findings are stated in bold throughout, numbered by chapter, and always for the conditions we tested. The main ones are these.

- **Flipping feature names hurts more than any shift of the data**: 0.20 AUROC, against 0.12 for reversing the shortcut and 0.02 (not significant) for covariate shift (Chapter 4).
- **Strategies that neutralise the shortcut help under its reversal more than under covariate shift** (interaction +0.11, Holm $p = 0.024$), but no strategy helps under flipped names (Chapter 4).
- **A trained selector that wins on a stand-in learner does not win on the model**; none of the ten pre-registered selector contrasts favours it (Chapter 5).

- **The answer forms in the last six of 28 layers, and the names decide it there**; the model's prior is decodable from layer 2, while the demonstrated rule hardly is (Chapter 6).

1.5 How to read this thesis

Chapter 2 reviews in-context learning, priors, shift and selection. Chapter 3 describes the testbed and the statistical protocol shared by every experiment. Part II holds the three experimental chapters. Each opens with the question left by the previous chapter and closes with its own limitations. Chapter 7 draws the results together, answers the anticipated criticism of each research question and sets out the limitations of the study as a whole, and Chapter 8 concludes. The appendices list the notation, the design details behind each claim, and how to reproduce every result.

CHAPTER 2

Background

This chapter reviews the five strands of work the thesis builds on: in-context learning and what it learns (section 2.1), the contest between priors and demonstrations (section 2.2), distribution shift and shortcuts (section 2.3), demonstration selection (section 2.4), and the tools for looking inside a model (section 2.5). It closes with the gap this thesis addresses (section 2.6). [2]

2.1 In-context learning

Brown et al. (2020) showed that a sufficiently large language model can perform a new task from a few demonstrations in its prompt, with no gradient updates; Dong et al. (2024) survey the area since. What the model learns from the demonstrations is contested. Min et al. (2022) found that replacing demonstration labels with random ones often costs surprisingly little, suggesting that demonstrations mainly convey the input distribution, the label space and the format. Pan et al. (2023) separate *task recognition*, retrieving a task the model already knows, from *task learning*, acquiring an input–label mapping from the demonstrations themselves. Xie et al. (2022) frame in-context learning as implicit Bayesian inference, in which demonstrations update a prior over latent concepts. These views matter here because a model that mostly recognises tasks will follow its priors whenever the demonstrations disagree with them.

Controlled studies train or probe models on synthetic function classes. Garg et al. (2022) show that transformers trained from scratch can learn linear functions and decision trees in context, and Zhang et al. (2024) find that such in-context learners of linear functions are brittle under covariate shift. Tabular foundation models are trained on synthetic task priors in the same spirit (Hollmann et al., 2023, 2025; Müller et al., 2022). For pretrained language models, tabular rows are serialised as text (Dinh et al.,

[2] **TODO** *Fold in the strongest material from the earlier literature review (Lit-review.pdf), and check each claim against the paper itself.*

2022; Hegselmann et al., 2023), and classification is read from the probabilities of the label tokens (Brown et al., 2020), which are affected by surface-form competition (Holtzman et al., 2021) and by a bias toward particular labels that contextual calibration corrects (Zhao et al., 2021). Public tabular datasets can be memorised by large models (Bordt et al., 2024), one reason to prefer synthetic tasks.

2.2 Priors against demonstrations

A model's prior is what it would predict with no demonstrations. When feature names carry meaning, the prior is semantic: a feature called "credit score" suggests a direction before any example is seen. Dinh et al. (2022) and Choi et al. (2022) show that feature names change what language models do with tabular data, and Nafar et al. (2025) compare named and anonymised features in in-context regression.

Wei et al. (2023) tested what happens when the demonstrations contradict the prior, by flipping labels or replacing them with semantically unrelated ones: only large models follow the flipped labels, and smaller models stay with their priors. At the scale studied here the picture is graded rather than all-or-nothing. Llama-3.1-8B-Instruct agrees with fully flipped sentiment labels about half the time (Krishna Kumar, 2025), and with flipped labels in-context accuracy plateaus below its level with the original labels (Kossen et al., 2024). Related work on knowledge conflicts studies when a model trusts its context over its parametric memory (Longpre et al., 2021; Xie et al., 2024); the statistical notion of prior–data conflict goes back further (Evans & Moshonov, 2006). This thesis treats the question as a contest between prior and demonstrations whose outcome can be shifted, and asks whether the demonstrations themselves can be chosen to shift it.

2.3 Distribution shift and shortcuts

Kinds of shift. Covariate shift changes the input distribution $P(X)$ while keeping $P(Y \mid X)$; label (prior-probability) shift changes $P(Y)$; concept shift changes $P(Y \mid X)$ itself (Moreno-Torres et al., 2012; Quiñonero-Candela et al., 2009; Shimodaira, 2000). On real tabular benchmarks, most of the large performance drops are dominated by concept shift (Liu et al., 2023). Benchmarks such as TableShift (Gardner et al., 2023) collect natural shifts, but each mixes several kinds, so the effect of one kind cannot be isolated.

Spurious features and shortcuts. A spurious feature predicts the label in the training environment for reasons that do not hold elsewhere. Colored MNIST pairs a noisy causal signal with a colour whose correlation with the label reverses at test time (Arjovsky et al., 2019), and Rosenfeld et al. (2021) model environmental features as class-conditional Gaussians. Learners adopt such shortcuts readily (Geirhos et al., 2020; Nagarajan et al., 2021). Worst-group robustness is measured over (label, spurious attribute) groups (Sagawa et al., 2020), and simple group balancing is a strong baseline (Idrissi et al., 2022). In-context learning inherits the problem: when demonstrations are consistent with several features, models show systematic preferences among them (Si et al., 2023), exploit shortcuts in the prompt (Tang et al., 2023), and in-context learners trained on tasks with spurious features adopt them (Chen et al., 2026; Harutyunyan et al., 2026). Restricting a predictor to causal features is not automatically better: on tabular shifts, Nastl and Hardt (2024) found predictors using all features generalised as well or better.

2.4 Demonstration selection

Heuristic selection. The simplest heuristics retrieve the training examples nearest to the query (Liu et al., 2022) or cover the input space with diverse examples (Gupta et al., 2023; Levy et al., 2023). Others target robustness: comparable demonstrations, minimally edited to flip the label, break spurious associations (Fan et al., 2024), and misconfidence-based selection prefers examples on which the model's output departs from the true label (Xu & Zhang, 2024). Label balance and order are controlled because they bias predictions (Lu et al., 2022; Zhao et al., 2021).

Learned selection. Several methods train a selector from the language model's own feedback. EPR (Rubin et al., 2022) and UDR (Li et al., 2023) train retrievers on the model's scores of candidate demonstrations, CEIL (Ye et al., 2023) learns to choose whole sets, and Zhang et al. (2022) learn a selection policy by reinforcement learning. Datamodels (Ilyas et al., 2022) regress a model's output on which training examples were present; Chang and Jia (2023) fit them to in-context examples on the correct-class margin, and Nguyen and Wong (2023) estimate influence from random subsets. Signals from model internals and learned judges have also been proposed (Kato et al., 2025; Wang et al., 2026); Kato et al. (2025) show that scores computed from induction-head representations track in-context accuracy but leave selection with them to future work.

Set-based architectures. The selector built in this thesis reads the whole pool of candidates at once. It draws on In-Context Risk Minimization (Gupta et al., 2024), in which a transformer adapts to an

environment by attending over other inputs from it, on the row tokens of TabPFN (Hollmann et al., 2023), and on attention over sets (Lee et al., 2019).

2.5 Looking inside the model

The residual stream of a transformer (Vaswani et al., 2017) carries a running representation that every layer reads from and writes to (Elhage et al., 2021). The *logit lens* reads an intermediate layer's prediction by applying the final normalisation and unembedding to it (nostalgebraist, 2020); its readings can be biased at early layers (Belrose et al., 2023). Halawi et al. (2024) used it to find a *critical layer* after which models given false demonstrations switch to the demonstrated labels. Linear probes test whether information is linearly decodable from a layer (Alain & Bengio, 2017), which shows that it is present, not that it is used (Belinkov, 2022). Steering adds a direction to the residual stream to change behaviour: activation addition (Turner et al., 2023) and contrastive activation addition (Rimsky et al., 2024) take differences of mean activations. Minder et al. (2025) find a single direction that controls whether a model follows its context or its prior, and task and function vectors (Hendel et al., 2023; Todd et al., 2024) compress an in-context task into one direction.

2.6 The gap

Three gaps meet in this thesis. First, selection methods are almost always evaluated in distribution, with no control over what kind of shift the queries undergo. Second, the conflict between semantic priors and demonstrations has been studied by flipping *labels*, not by changing what the *features* mean, and not under shift. Third, the learned selectors above are trained on the model's scores of whole prompts; none, as far as our search reached, uses the model's measured prior on each candidate, and none is tested on whether it helps a small model overcome that prior. The nearest work, Kato et al. (2025), is correlational and builds no selector. This thesis builds the testbed that the first two gaps require, and uses it to test the third.

CHAPTER 3

Method: a controlled testbed

Every experiment in Part II runs on the same testbed: synthetic tasks with known structure, three environments that each change one thing, a naming factor that sets the model's prior against the rule, a fixed prompt and scoring procedure, and a shared statistical protocol. This chapter describes each in turn. Every symbol is listed in Appendix A, and the full definitions, with proofs of the properties stated here, are in Appendix B.

3.1 Why synthetic tasks

The project began on real tabular benchmarks. TableShift (Gardner et al., 2023) names covariate, label and concept shift, but every one of its shifts mixes them, so a drop in accuracy cannot be attributed to one kind. Real data also give no ground truth for which features determine the label, which **RQ3** needs, and public datasets may have been memorised (Bordt et al., 2024). Synthetic tasks solve all three problems, in the tradition of Garg et al. (2022) and of the synthetic priors behind tabular foundation models (Hollmann et al., 2023; Müller et al., 2022).

3.2 Tasks

Each task has ten features. Three are *load-bearing*: the label depends on them and on nothing else. Five are *distractors*. One, f_8 in the generator's numbering, is a *spurious shortcut*, and one, f_9, is pure noise. The ten features are shown in a random order recorded per task, so no feature's role can be read from its position.

Rules. Half of the tasks are **linear**: $P(y = 1 \mid x) = \sigma(c^\top x_C)$ over the load-bearing features C, with coefficients $c_j = u_j r_j$, $u_j \sim \mathrm{U}(1, 2)$ and a random sign r_j, so every load-bearing feature has a definite

direction. The other half are **trees**: a signed majority vote over the three features' signs, which makes them monotone in each feature as well. Labels are then flipped with probability $\eta = 0.02$.

The shortcut. The spurious feature is generated from the clean label,

$$x_8 = \sigma_t\big((2y_{\text{clean}} - 1)\,\Phi^{-1}(s) + \varepsilon\big), \qquad \varepsilon \sim \mathcal{N}(0, 1),$$

so its sign agrees with the label with probability $s \sim \mathrm{U}(0.80, 0.90)$, the range of colour–label agreement in Colored MNIST's training environments (Arjovsky et al., 2019). On its own it predicts the label better than any single rule feature. The spurious direction $\sigma_t \in \{+1, -1\}$ is drawn per task and balanced, for a reason that only appeared once the model was run: Qwen's zero-shot answers rise with every feature value, so with σ_t fixed at $+1$ that habit backed the shortcut in every task (section 4.2).

Suites. The evaluation suite has 24 tasks: 12 linear and 12 tree, crossed with two domains (loan approval and medical risk) and balanced on σ_t. A separate 12-task pilot suite was used to choose the number of demonstrations, so that choice never touched the evaluation tasks. Each task has a 256-row demonstration pool and, per environment, a label-balanced set of 20 queries. All features are standardised with the pool's statistics. The selectors of Chapter 5 are trained and validated on further tasks from the same distribution with disjoint seeds.

3.3 Environments

Queries come from three environments; the demonstration pool is always in distribution.

In distribution (ID):

Queries are drawn exactly like the pool.

Covariate shift:

Two load-bearing features are moved by 1–2 standard deviations in opposite label directions, with equal pushes on the label, and one distractor is moved as well. $P(y \mid x)$ is unchanged, and so is the label rate, by construction. An earlier design filtered random shifts on the label rate; it turned out to accept almost only shifts of distractors, which cannot affect any label (21 of 24 tasks).

Spurious reversal:

The shortcut's sign is reversed: it now agrees with the label in only $1 - s$ of rows.

Concept shift is realised by naming (next section), because changing $P(y \mid x)$ itself while keeping the task recognisable is exactly what flipped names do from the model's point of view.

3.4 Naming: setting the prior against the rule

Each task is shown under three namings. **Abstract** names are $f_0, \ldots, f_9$. **Aligned** names give each load-bearing feature a domain name whose everyday meaning agrees with its direction in the rule: in the loan domain, a feature that raises approval might be called "credit score". **Flipped** names give the same feature the antonym, "missed payments". Aligned and flipped prompts differ only in the three load-bearing names; non-causal features get plausible names with weak priors ("form version", "office floor number"), so names do not reveal which features matter.

Antonym pairs were screened on the model itself. On 2,000 zero-shot profiles, a numeric prior dominated: every abstract feature had a positive slope, and names that should lower $P(y = 1)$ mostly just raised it less. Only 1 of 16 candidate pairs met the planned symmetry rule. The final names are the five pairs per domain with the largest *measured* contrast. On the pilot tasks, zero-shot AUROC is 0.594 with aligned names and 0.430 with flipped names ($\Delta = 0.164$, 95% CI [0.053, 0.287], 10 of 12 tasks), so the names do move the prior; the effect is large for loan (0.28) and weak for medical (0.05).

3.5 Prompts, scoring and the model

Rows are serialised as `name: value` pairs separated by semicolons, followed by `->` and the label. The system message states that each example lists ten standardised measurements and a label, that the label follows an unknown rule over some of them, and that the model should answer with one character, 0 or 1. An abstract-name prompt reads, in part:

```
f0: 0.99; f1: 1.42; f2: 0.72; f3: -0.23; f4: -0.52; f5:
1.09; f6: 0.35; f7: -0.74; f8: -0.99; f9: -1.63 -> 0
...
f0: 1.25; f1: 0.19; f2: 0.10; f3: -0.04; f4: -0.83; f5:
0.20; f6: 0.01; f7: 0.36; f8: -0.31; f9: 0.65 ->
```

The model is **Qwen2.5-7B-Instruct** (Qwen et al., 2024) in bf16, run through PyTorch and Hugging Face `transformers` (Paszke et al., 2019; Wolf et al., 2020). Its score for a query is the margin

$m = \log p(\texttt{1}) - \log p(\texttt{0})$ of the two label tokens, computed in fp32 from the final hidden state; reading it from the bf16 output layer rounded margins onto a coarse grid. The demonstration prefix is computed once and cached, and each query is scored against the cached keys and values (Gim et al., 2024; Pope et al., 2023); this was checked to change mean per-task AUROC by at most 0.005. Qwen2.5-7B (base) answers in format and is used for exploratory comparisons. Inference ran first on an Apple M3 Max and then on a rented NVIDIA H200 with eight copies of the model sharing the GPU (the main grid, about 255,000 forward passes, took 42 minutes); the whole project used a few tens of US dollars of GPU time, which bounds what could be tested (section 7.4).[3] Llama-3.1-8B-Instruct (Grattafiori et al., 2024) was dropped after the pilot, because with this prompt it explains its reasoning instead of answering (only about 2% of its probability fell on the label tokens).

The number of demonstrations. On the pilot suite, $k = 8$ gold demonstrations beat the same rows with shuffled labels by $\Delta\text{AUROC} = 0.098$ (95% CI [0.021, 0.188], 9 of 12 tasks), so the model does use the labels; $k = 32$ did not pass. All experiments use $k = 8$.

Measured priors. The model's prior on a row is its zero-shot margin on that row, under the unit's naming, centred on the pool median. A ridge surrogate of this prior from feature names (Hoerl & Kennard, 1970) failed its planned fit criterion (held-out R^2 0.20–0.51 against a bar of 0.5), so selection uses the measured margins directly.

3.6 Selection strategies

Table 3.1 lists the strategies. Every strategy except `random` and `similarity` takes $k/2$ demonstrations of each label, and balanced strategies are compared with `label_diversity`, a label-balanced random draw, so that contrasts compare selection mechanisms at fixed label counts.

3.7 Evaluation and statistics

The metric. The primary statistic is AUROC (Fawcett, 2006; Hanley & McNeil, 1982) of the margin, computed within each cell of (task, strategy, seed, naming, environment) over its 20 label-balanced queries. It was chosen over calibrated balanced accuracy, after the first tasks had run and before any

[3] **TODO** *State the total cloud cost from the Runpod billing history.*

TABLE 3.1. Selection strategies. Ground-truth strategies (marked GT) see feature roles that a deployed selector would not; this is disclosed wherever they are compared with SATA.

Strategy	What it selects
`zero_shot`	No demonstrations.
`random`	Uniform draw from the pool.
`label_diversity`	Uniform draw, $k/2$ rows of each label: the comparator.
`feature_range`	Covers the quantile bins of the three features with the highest mutual information with the label.
`rule_diversity` (GT)	Covers the ground-truth regimes of the rule, one row each.
`counter_spurious` (GT)	Matched pairs on $\lvert x_8 \rvert$ in which the shortcut agrees and disagrees equally often, so it carries no information.
`similarity`	The $k/2$ nearest rows of each label to the query in feature space (Liu et al., 2022).
`counter_prior`	Clean rows whose label contradicts the model's measured prior, when the pool shows prior–data conflict; related to misconfidence-based selection (Xu & Zhang, 2024).
SATA, SATA-MI	Learned selectors (Chapter 5).

contrast was computed: contextual calibration left every query of a cell on one side of the threshold in 36% of plain-random cells, which pins balanced accuracy at 0.5 and forces shift gaps to zero. Within-cell AUROC has a blind spot of its own, which section 4.2 measures: a shift that moves every score together cannot be seen.

Inference. Demonstration sets depend only on (task, strategy, seed), so the same sets are reused across namings and environments, and contrasts are paired. Uncertainty comes from a **hierarchical bootstrap** (Davison & Hinkley, 1997; Efron & Tibshirani, 1993): each of 2,000 draws resamples the 24 tasks and then, within each task, the positive and negative queries separately. Contrasts are one-sided, and each family is **Holm-corrected** (Holm, 1979). Every confirmatory family was **written into the code before the data it tests were seen** (Nosek et al., 2018): ten contrasts for **RQ1**–**RQ3**, six for SATA and four for SATA-MI. Everything outside those families is labelled exploratory.

Figures. Following Healy (2019), estimates are positions on a common scale with 95% intervals, categories are ordered by value, and colour carries one meaning per figure, in colour-blind-safe palettes; every figure is drawn from the result files by one script.

Gates. Each expensive stage was preceded by a check that the instrument worked (cache exactness, answer format, learnability, naming strength, and an oracle check of every selector's targets), because an earlier, ungated version of this project produced results that turned out to be artefacts of code defects.

CHAPTER 4

Shift, names and the prior

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*Research questions: **RQ1**, **RQ2** and the behavioural half of **RQ3**.*

4.1 Motivation

Before asking whether a selector can help, we need to know what goes wrong: how each shift affects Qwen2.5-7B-Instruct, whether any hand-designed strategy helps under the shift it was designed for, and which features the model relies on. First, what the pilot taught us, because it changed the design of everything after it.

4.2 What the pilot revealed: a numeric prior

In the first pilot run the shortcut's direction was the same in every task: a larger x_8 always meant label 1. With abstract names, Qwen's zero-shot margin rose with every feature value (Spearman correlation 0.70 with the plain sum of the ten features). Scoring the ID queries by that sum gives AUROC 0.628, and 0.511 once x_8 is left out; Qwen's own zero-shot AUROC was 0.633. Its "zero-shot skill" was the shortcut, read through a habit that bigger numbers mean 1.

Finding 4.1. Qwen2.5-7B-Instruct carries a numeric prior, "bigger values mean 1", that applies to every feature, including the shortcut.

We made the shortcut's direction random per task and balanced it, and reran the pilot on the same tasks. Zero-shot AUROC fell to 0.499 overall: 0.642 where the habit backs the shortcut ($\sigma_t = +1$) and 0.357 where it opposes it ($\sigma_t = -1$). The tasks whose shortcut had not changed gave bit-identical scores. From here on, every analysis of the shortcut reports the two halves separately as well as pooled.

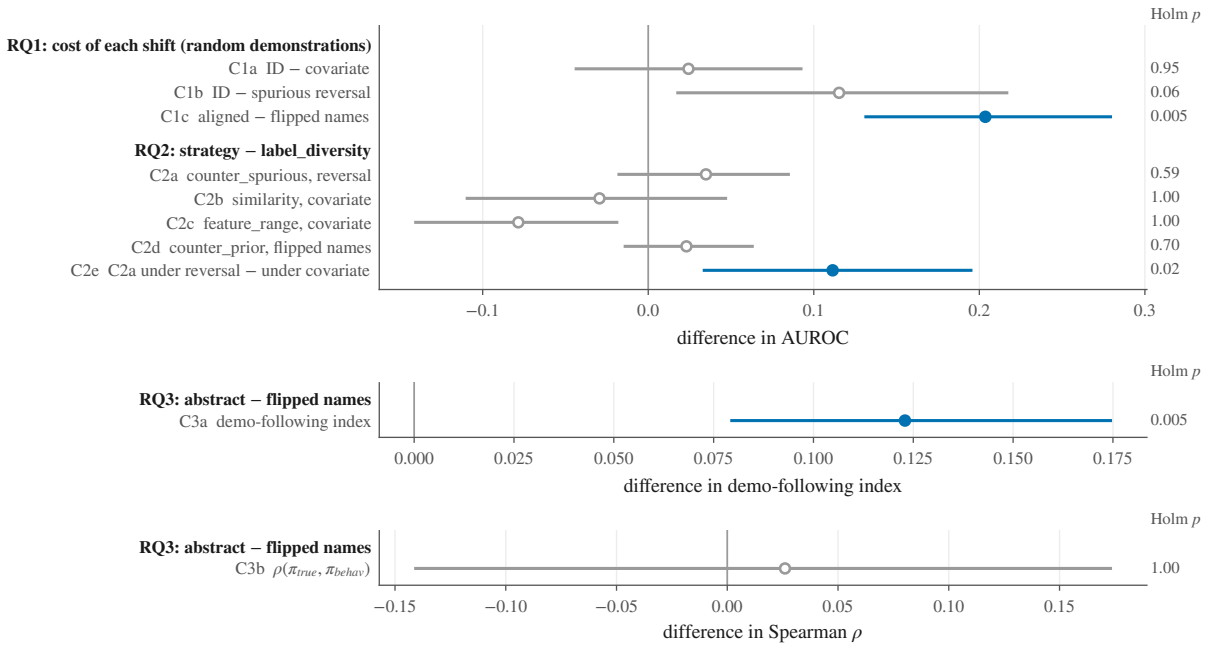


FIGURE 4.1. The ten confirmatory contrasts for **RQ1**–**RQ3**. Each row is an estimate with its 95% interval (hierarchical bootstrap over 24 tasks); filled blue marks the three that survive Holm correction (adjusted p at the right), hollow grey the rest. The three panels use different units, so they do not share a scale. Concept shift through naming (C1c) is the largest effect.

4.3 Setup

The main grid crosses the 24 evaluation tasks, three namings, three environments and every strategy of Table 3.1 except the learned selectors, with three demonstration seeds (one for `similarity`, whose nearest-neighbour sets barely change with the seed). It was run on Qwen2.5-7B-Instruct; a reduced grid on Qwen2.5-7B (base) used abstract and flipped names. The confirmatory family of ten contrasts (Figure 4.1; exact values in Table B.1) was frozen before any result.

4.4 RQ1: how much does each shift hurt?

Finding 4.2. Flipping the feature names costs more than any shift of the data: 0.20 AUROC for random demonstrations on ID queries (95% CI [0.13, 0.28], 22 of 24 tasks).

Reversing the shortcut costs about half as much (C1b, +0.115), and its interval excludes zero but it misses significance after correction (Holm $p = 0.063$). Covariate shift shows no cost in within-cell

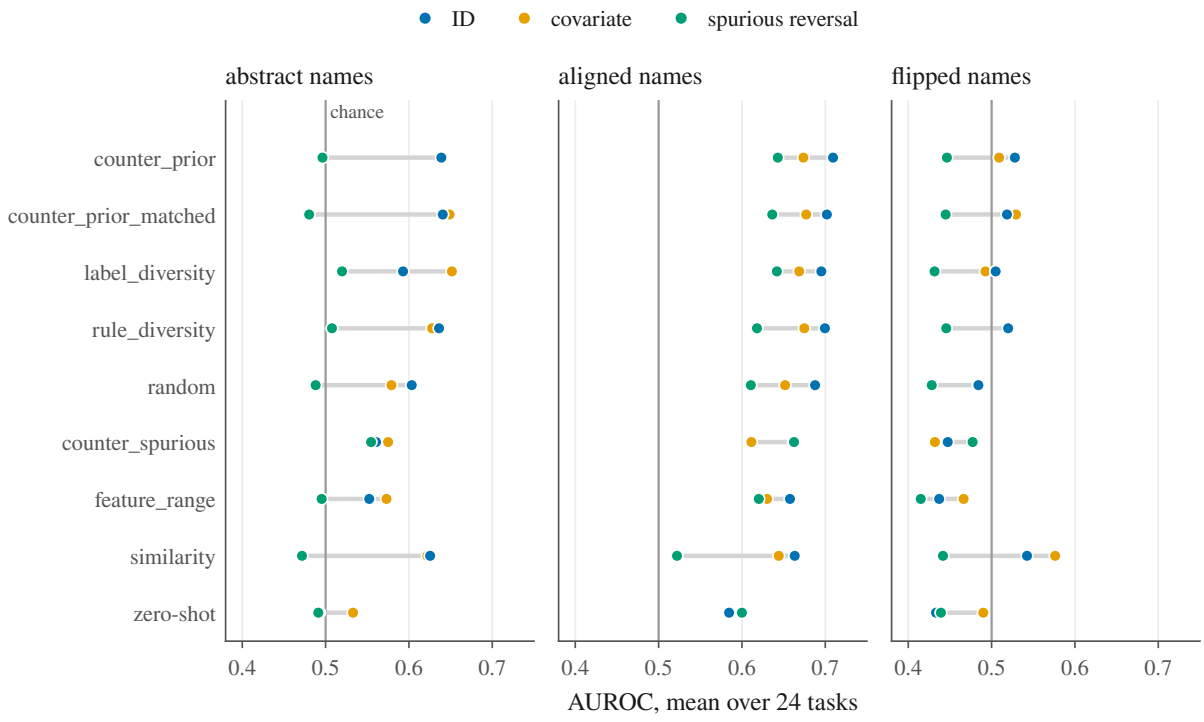


FIGURE 4.2. Every strategy by naming and environment (Qwen2.5-7B-Instruct, mean AUROC over 24 tasks). Each row joins one strategy's three environments; rows are ordered by mean AUROC under aligned names. The panels differ far more than the rows within a panel: the names matter more than the choice of demonstrations.

AUROC (C1a, +0.024). Figure 4.2 shows the size of the naming effect for every strategy: under flipped names every strategy stays near chance on ID queries (AUROC 0.44–0.54), and aligned names lift every strategy to 0.66–0.71.

The covariate null is a measurement limit. A null on covariate shift is not evidence that covariate shift is harmless. The shift raises every covariate query's score together, by 0.44 to 1.21 logits for the same prompt; ranking ID positives against covariate negatives gives AUROC 0.46–0.54, but within-environment AUROC cannot see a uniform move. An exploratory variance shift, which stretches every covariate but the shortcut by 1.5–3 times, does hurt, but only under flipped names (AUROC -0.09 to -0.13).

Finding 4.3. Within-environment AUROC hides covariate harm that moves every score together; covariate shift is not shown to be harmless.

4.5 RQ2: does the right strategy help under the right shift?

Each strategy was designed for one kind of shift: `counter_spurious` against the shortcut, `similarity` and `feature_range` against covariate shift, `counter_prior` against prior conflict. Individually, none beats `label_diversity` after correction (C2a–C2d). `feature_range` is significantly worse under covariate shift by its interval (-0.078); it turned out to cover the shortcut in all 24 tasks, because the shortcut is the most label-related feature in every pool.

Finding 4.4. Neutralising the shortcut helps under its reversal more than under covariate shift (interaction +0.111, Holm $p = 0.024$, 17 of 24 tasks).

The interaction is the only evidence in the family that a strategy's value depends on the kind of shift. `counter_prior`, the strategy aimed at the prior, does not help under flipped names (C2d, +0.023); in nine tasks the pool shows too little prior–data conflict for it to activate.

4.6 RQ3: which features does the model rely on?

On `label_diversity`'s sets and ID queries, π_{true} is each feature's ground-truth importance, π_{behav} the drop in correct-label margin when the feature is replaced by another pool row's value (which keeps the query in distribution; Andridge and Little, 2010; Zhu et al., 2026), and π_{self} the model's own ranking. The *demo-following index* nudges each load-bearing feature by ± 0.5 and records how often the margin moves in the rule's direction rather than the name's.

Finding 4.5. With abstract names, Qwen relies on the shortcut, not the rule: replacing the shortcut costs 0.141 of correct-label margin, and the rule features cost 0.009.

With aligned names the rule features dominate (0.241). With flipped names the model uses them in the names' direction (-0.047), and its demonstration-following falls (C3a, +0.123 lower than under abstract names, 23 of 24 tasks). The correctness of its reliance does not change significantly between namings (C3b). [4]

Finding 4.6. Qwen's self-reports are unfaithful to its behaviour.

[4] **TODO** *Add the demo-following index for each naming.*

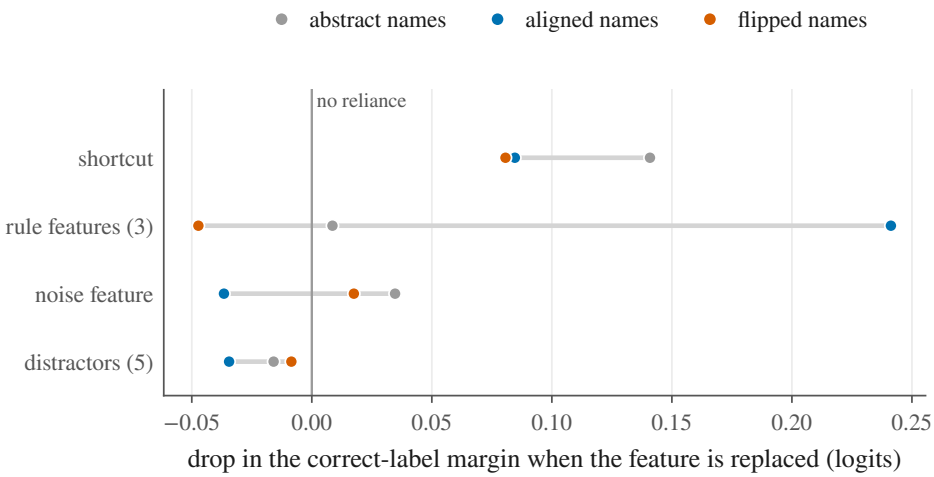


FIGURE 4.3. Which features Qwen relies on (π_{behav}, the drop in correct-label margin when a feature is replaced by a value from another row), by naming. Under abstract names the shortcut carries the reliance; under aligned names the rule features do; under flipped names the rule features are used in the names' direction (negative).

All 216 rankings parsed. With names, Qwen ranks the named rule features in its top three (0.64–0.69 of rankings) even under flipped names, and the rank correlation between what it reports and what it relies on is -0.02 (abstract), 0.12 (aligned) and -0.02 (flipped), in line with work on unfaithful self-explanations (Lanham et al., 2023; Turpin et al., 2023).

4.7 The prior reads the shortcut in the query

Splitting the cost of spurious reversal by the shortcut's direction explains much of C1b (Figure 4.4). Where the "bigger means 1" habit agrees with the shortcut ($\sigma_t = +1$), reversal costs every strategy 0.04–0.21 AUROC. Where it opposes the shortcut ($\sigma_t = -1$), most gaps are near zero or negative. Zero-shot, with no demonstrations at all, splits the same way: $+0.07$ to $+0.13$ against -0.09 to -0.15.

Demonstrations without the shortcut. If demonstrations are meant to teach the causal rule, perhaps they should not show the shortcut at all. This exploratory experiment removed the shortcut (and, in a second arm, also the noise feature) from every demonstration while keeping queries complete. Removing the shortcut cut ID AUROC by 0.048 and covariate AUROC by 0.040, and raised spurious-reversal AUROC by 0.049. The ID–reversal gap shrank in all 18 (strategy, naming) cells, significantly in 14. Removing the shortcut worked as well as neutralising it with `counter_spurious`, and removing the

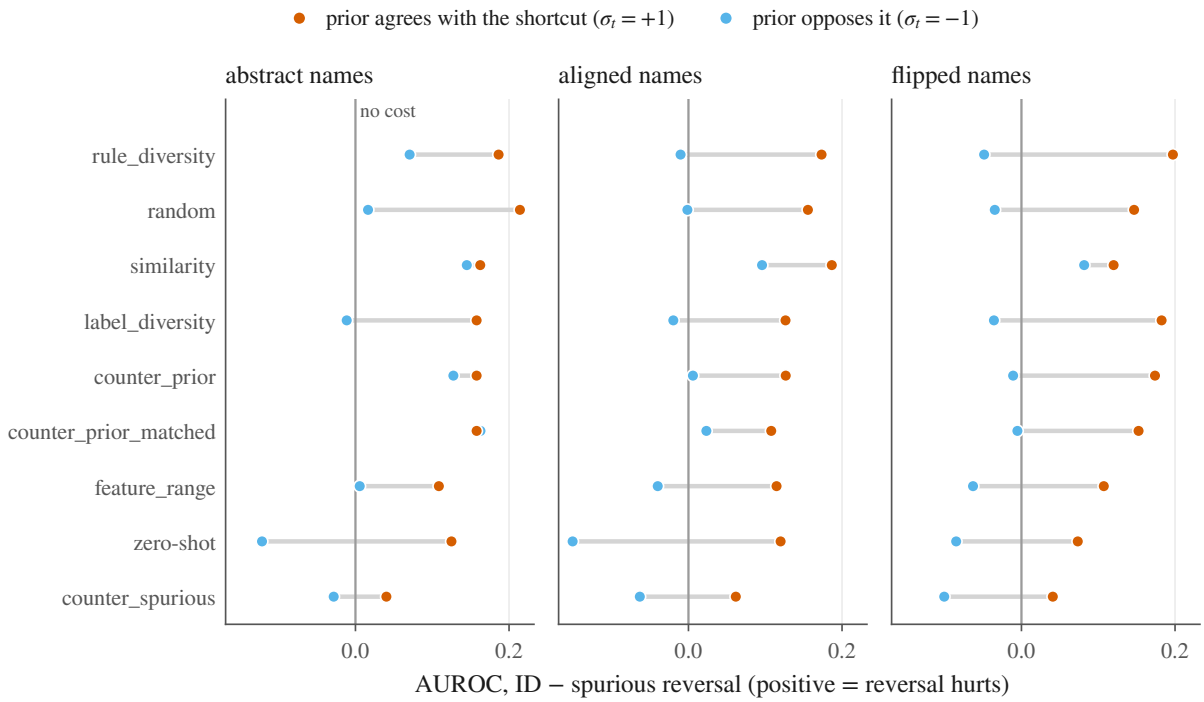


FIGURE 4.4. The cost of spurious reversal (ID minus reversal AUROC), split by whether Qwen's "bigger means 1" prior agrees with the shortcut ($\sigma_t = +1$, vermilion) or opposes it ($\sigma_t = -1$, blue). The cost appears where the prior agrees with the shortcut, with or without demonstrations (zero-shot row).

noise feature added nothing. The gap that remains follows σ_t ($+0.04$ to $+0.11$ against -0.06 to -0.15): the prior still reads the shortcut in the query.

Finding 4.7. Much of what looks like shortcut learning from demonstrations is the model's numeric prior reading the shortcut in the query; hiding the shortcut from the demonstrations removes the demonstration part and leaves the prior part.

The base model. Qwen2.5-7B (base) loses less to flipped names than the instruction-tuned model (random demonstrations on ID queries: 0.625 to 0.560, against 0.603 to 0.484), which matters in Chapter 5, where it is the one model a learned selector helps.

4.8 What this chapter does not show, and the next question

Every strategy tested reads only the data, never what the model makes of it (`counter_prior` is a single-rule heuristic, inactive in nine tasks), and only one interaction (C2e) shows that matching strategy

to shift pays; with 24 tasks, effects below about 0.05 AUROC are hard to detect. The natural next step is to *learn* a selector that composes these ideas, choosing per query the rows that teach the rule rather than the shortcut: Chapter 5.

CHAPTER 5

Learning to select

Research question: ***RQ4*** *(does a learned selector beat the hand-designed strategies, and do signals measured from the model help it?).*

5.1 Motivation

Chapter 4 left an opening. Neutralising the shortcut helped under its reversal (C2e), but the strategies that did so were hand-built rules for one shift each. A learned selector could combine their ideas, choosing per query the rows that teach the rule rather than the shortcut, and could be trained on synthetic tasks where the best choices are known. This chapter builds one, SATA (section 5.3), and then gives it what the model itself believes about every row, SATA-MI (section 5.4). Each was tested against a contrast family frozen before its first result on the model.

5.2 The selector

SATA scores every row of the 256-row pool given the whole pool and the query. Each candidate is one token: its feature vector plus a label embedding plus a "demonstration" type embedding; the query is a token with its feature vector and a "query" type embedding. A transformer encoder ($d = 128$, 4 heads, 4 layers, about 0.8M parameters) attends over all tokens, and a linear head gives one score per candidate, normalised by a softmax. There are no positional encodings, so the encoder is permutation-equivariant over candidates (Lee et al., 2019); the row encoding resembles TabPFN's (Hollmann et al., 2023), and the idea of adapting by attending over other inputs from the environment comes from Gupta et al. (2024). Unlike EPR, UDR or CEIL (Li et al., 2023; Rubin et al., 2022; Ye et al., 2023), SATA is not trained on the language model's scores: it learns targets computed from the ground truth, so its training never touches the model it is evaluated on.

Getting the targets right. An oracle check required the targets' own choices to beat every strategy on a stand-in learner: gradient-boosted trees (Chen & Guestrin, 2016) fitted on the eight chosen rows and scored on the queries. The targets as first designed, which rewarded shortcut-breaking rows, failed: they favoured label-noise rows (whose shortcut follows the true label, so they look counter-spurious) and, without those, reversed the shortcut (it agreed in only 25% of chosen rows). The target kept is each row's probability of appearing in a `counter_spurious`-style matched set: within each label, two clean rows that break the shortcut, each paired with the clean agreeing row nearest in $|x_8|$ (Appendix B). Its oracle reaches a worst environment of 0.640 against 0.555 for the best strategy.

Drawing instead of ranking. A learned score ranks shortcut-breaking rows first but cannot pin down which agreeing row is each one's partner, so taking the top k reversed the shortcut again. Since the scores are trained as a distribution over rows (with a KL loss), SATA *draws* $k/2$ rows of each label in proportion to them: 0.558 against 0.478 for top-k on 50 validation tasks.

Finding 5.1. A selector's training target can encode the very shortcut it is meant to avoid, and an oracle check on a stand-in learner catches it before any training.

5.3 SATA on the model

SATA was trained on 1,000 tasks from the evaluation distribution, regenerated every epoch with a fresh permutation of feature positions so that the shortcut's position carries no information, and validated on 100 more (Adam (Kingma & Ba, 2015) at learning rate 10^{-4}, early stopping (Prechelt, 1998)). On the stand-in learner its worst environment is 0.652, against 0.555 for the best hand-designed strategy, with no weak environment (Figure 5.1).

Finding 5.2. On the stand-in learner, SATA beats every hand-designed strategy in every environment.

The **RQ4** family (R1–R6; Figure 5.2, exact values in Table B.1) was frozen before any SATA result on the model.

Finding 5.3. On Qwen2.5-7B-Instruct, SATA does not beat balanced random demonstrations: none of the six pre-registered contrasts favours it.

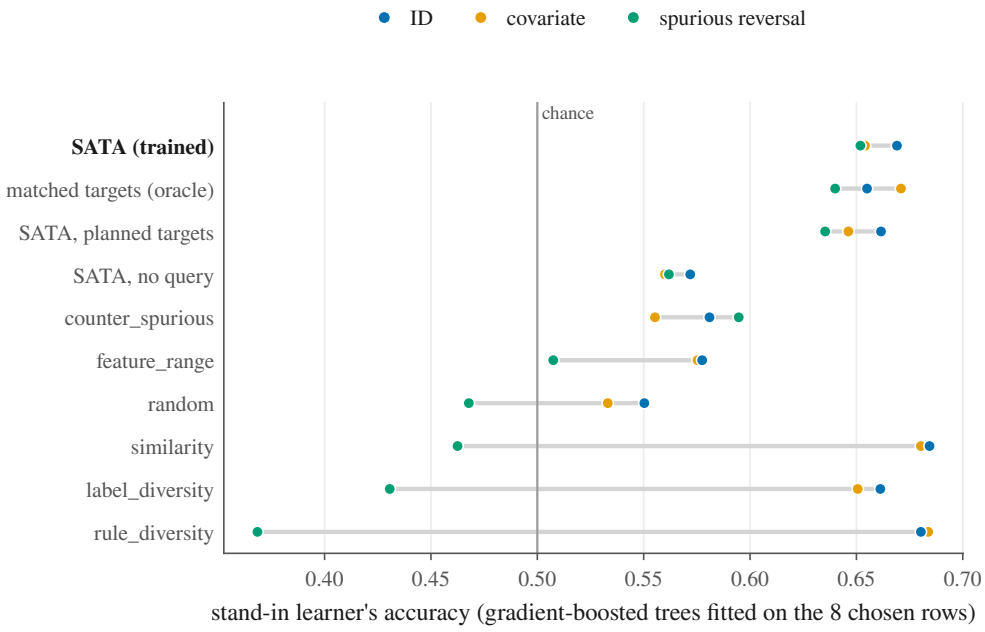


FIGURE 5.1. The stand-in learner's accuracy for each selection method, per environment (100 validation tasks), ordered by worst environment. Trained SATA matches its oracle targets and has no weak environment; the hand-designed strategies trade one environment against another.

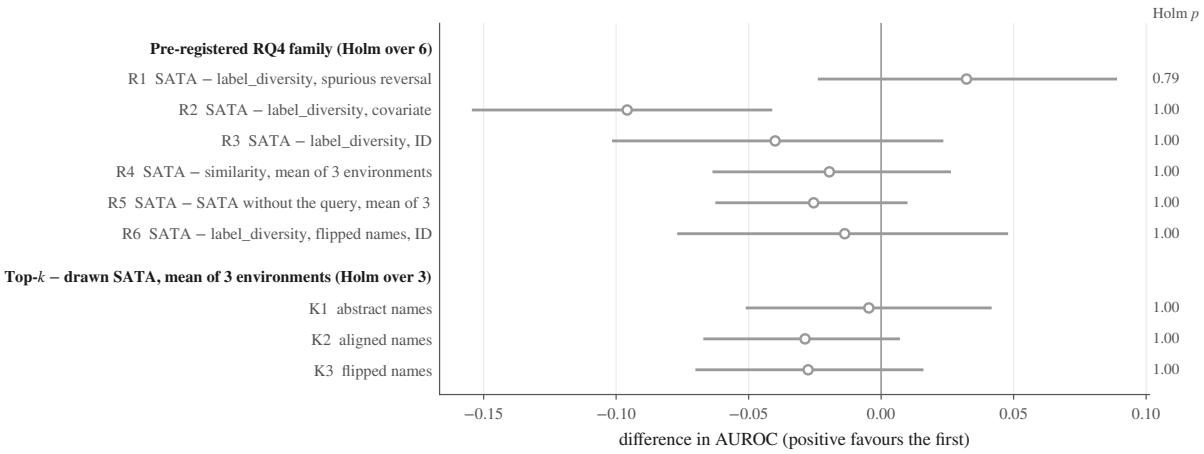


FIGURE 5.2. SATA on Qwen2.5-7B-Instruct: the pre-registered **RQ4** family (top) and the check of drawing against top-k selection (bottom), with 95% intervals over 24 tasks. No contrast survives correction (all hollow); R2's interval lies below zero.

SATA flattens the environments rather than raising them. Under abstract names its ID, covariate and reversal AUROC are 0.553, 0.556 and 0.552, the same pattern as `counter_spurious` (0.561, 0.575, 0.555): a gain under reversal paid for on ID and, significantly, under covariate shift (R2, -0.096 $[-0.154, -0.041]$).

Was it the drawing? Running SATA's top-k checkpoint with top-k selection on the same tasks does no better: top-k minus drawn SATA is -0.005, -0.029 and -0.028 under abstract, aligned and flipped names, every interval crossing zero (Figure 5.2, bottom).

The base model is different. On Qwen2.5-7B (base), in an exploratory comparison without Holm correction, SATA has the best worst environment of any strategy under both namings: 0.589 under abstract names (`counter_spurious` 0.550, `random` 0.504) and 0.551 under flipped names (0.471 and 0.461), gaining $+0.085$ and $+0.094$ over random demonstrations under spurious reversal, at no ID cost.

Finding 5.4. The same selector that ties on the instruction-tuned model gives the base model, whose priors pull less, the best worst-case AUROC of any strategy (exploratory).

5.4 SATA-MI: telling the selector what the model believes

SATA chose good rule evidence, but the stand-in learner that rewarded it ranks strategies like Qwen only moderately (Spearman 0.68 under abstract names, 0.46 under named ones), and Chapter 4 showed why: Qwen-Instruct reads the shortcut and its numeric prior. If the selector knew, for every candidate and for the query, what the model predicts without demonstrations, it could choose rows that work against that prediction where it is wrong.

Design. SATA-MI reads, for every pool row and the query, its feature vector and Qwen's measured zero-shot margin m_0 under the task's naming, centred on the pool median. Its controls are `sata_names`, which reads $s^{\text{lex}} \odot x$ (each name's dictionary direction, no model measurement) instead, and `counter_prior`; both selectors learn SATA's targets on 300 tasks. Because gradient-boosted trees cannot see names, a new stand-in, a *reader*, scores a set by how Qwen combines evidence with its prior: $P(\text{Qwen says } 1) = \sigma(a_1\, \hat{w}^\top x + a_2\, m_0 + a_0)$, with $\hat{w}$ a ridge logistic fit on the eight chosen rows and a_0, a_1, a_2 fitted once on Qwen's predictions. A gate required the reader to rank strategies like Qwen (Spearman ≥ 0.5) and SATA-MI to beat both controls; the family R7–R10, under flipped names, was written down before any result; training was capped at 30 epochs for budget.

Results. On the reader, SATA-MI reached 0.620, its targets' own level, against 0.590 for `sata_names`, all of the gap under spurious reversal. But the gate failed: the reader agreed

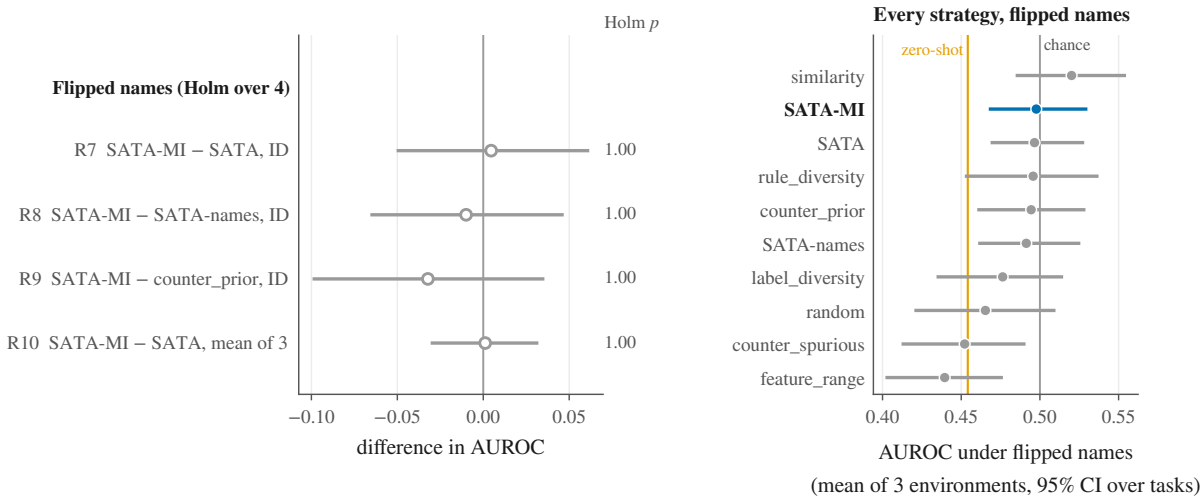


FIGURE 5.3. SATA-MI under flipped names. Left: the pre-registered family R7–R10, with 95% intervals; none is significant. Right: every strategy's AUROC (mean of the three environments, 95% interval over tasks), with SATA-MI highlighted and zero-shot marked; every strategy sits near chance.

with Qwen at Spearman 0.60 under abstract names and only 0.48 and 0.47 under aligned and flipped names, and SATA-MI was below `sata_names` on the validation tasks under named features.

Finding 5.5. On Qwen2.5-7B-Instruct, reading the model's prior changes nothing: all four pre-registered contrasts are null (Holm $p = 1.00$).

Under flipped names every strategy stays near chance (means 0.44–0.52; zero-shot 0.454), SATA-MI included (Figure 5.3). No choice of eight demonstrations moves the model far from its prior when the names point the wrong way.

5.5 What this chapter does not show, and the next question

Both stand-in learners tracked the model only moderately under named features, so both selectors were optimised for a learner that is not Qwen; and SATA-MI's 30-epoch cap may have left a little on the table, though it had already matched its own targets. The larger limitation is that behaviour alone cannot say *why* no choice of evidence works. Two selectors, one reading only the data and one reading the model's prior, both helped a stand-in and both failed on the instruction-tuned model. Either the demonstrated rule never reaches the computation that produces the answer, or it reaches it and loses there. Chapter 6 looks inside the model to tell these apart.

CHAPTER 6

Inside the model

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*Research question: **RQ3**, second half (why does the prior win?).*

6.1 Motivation

Three chapters of behaviour converge on one picture: Qwen2.5-7B-Instruct follows its prior and the feature names, and no selection strategy, hand-designed or learned, changes that. This chapter asks where in the model that happens. It asks at which layer the answer starts to depend on the names, whether any selection strategy moves that layer, whether the demonstrated rule is present in the model's internal state at all, and whether a single direction in that state makes the model follow the demonstrations.

6.2 Method

Logit lens. At every layer l of 28, the last-position residual state h_l is passed through the model's final normalisation and the difference of the two label rows of the unembedding, $w = W_U[1] - W_U[0]$, giving a per-layer margin $m_l = w^\top \mathrm{norm}(h_l)$ (Elhage et al., 2021; nostalgebraist, 2020). At the last layer this is exactly the margin of section 3.5; this check holds within 8.6×10^{-6} on all 25,920 rows, against a bar of 10^{-3}. The probes reuse the demonstration sets and prompts of the earlier chapters unchanged: the 24 evaluation tasks under three namings, seed 0, for zero-shot, `random`, `label_diversity`, `counter_prior`, SATA and SATA-MI.

Critical layer. Following Halawi et al. (2024), the critical layer is the first layer at which $\mathrm{AUROC}_l(\text{aligned}) - \mathrm{AUROC}_l(\text{flipped})$ exceeds 0.1 with a task-bootstrap interval above zero. We define it with AUROC rather than the accuracy of $\mathbb{1}[m_l > 0]$, because Qwen's bias toward label 1 would

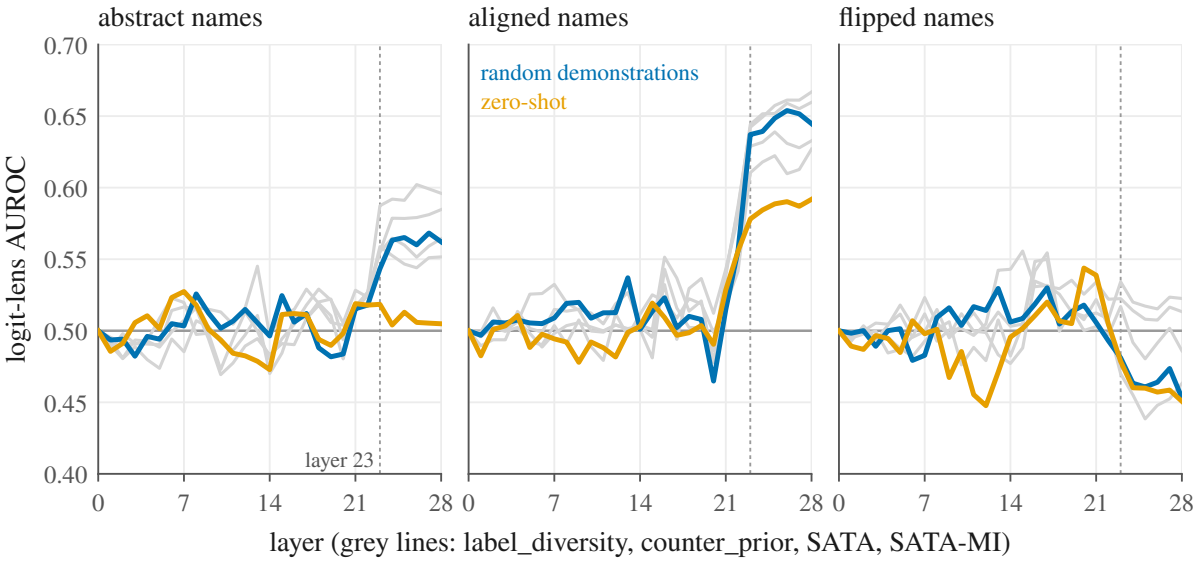


FIGURE 6.1. Logit-lens AUROC against the rule label at every layer (24 tasks, seed 0). Random demonstrations and zero-shot are highlighted; the four other strategies are in grey. The read-out is at chance for 20 layers and splits by naming after layer 22 (dotted line: the critical layer, 23).

otherwise decide the accuracy (section 3.7); accuracy of the median-centred prediction is reported alongside.

Probes. Logistic probes on h_l (Alain & Bengio, 2017), trained on some tasks and tested on others (four folds by task), decode either the rule label or Qwen's own zero-shot prediction, with random demonstrations.

Steering. On queries where the rule label and Qwen's zero-shot prediction disagree, v_l is the mean residual of predictions that follow the demonstrations minus the mean of those that follow the prior, averaged over the two rule labels, in the manner of activation addition (Rimsky et al., 2024; Turner et al., 2023). It is fitted on the 12 even-indexed tasks and applied, as $h_l \leftarrow h_l + \alpha v_l$ during the query only, on the 12 odd-indexed tasks at layers 8, 12, 16, 20 and 24, with α from 1 to 32 and a random direction of the same norm as a control.

6.3 Results

Finding 6.1. Qwen's answer forms in its last six of 28 layers; for the first 20 its label read-out is at chance under every naming and strategy.

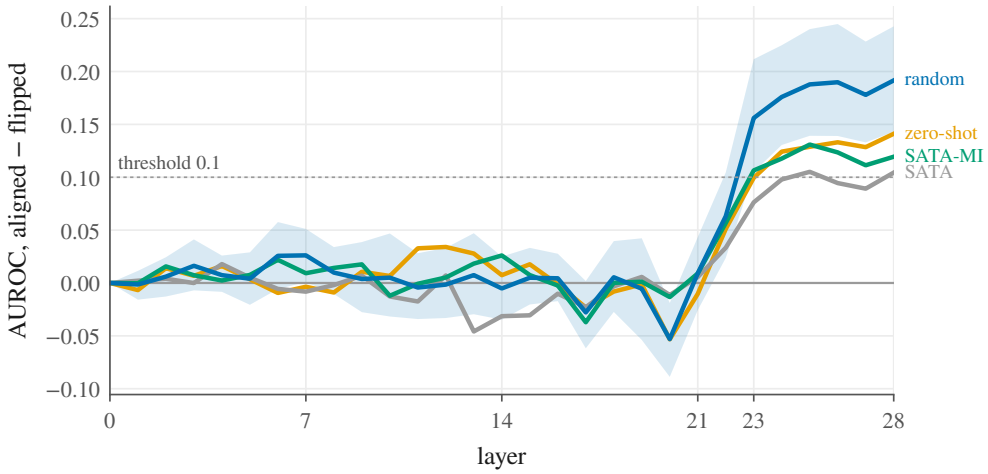


FIGURE 6.2. Aligned minus flipped AUROC by layer, with a 95% band over tasks for random demonstrations. The gap crosses the 0.1 threshold at layer 23 for every strategy, zero-shot included, so the names, not the demonstrations, drive it.

From layer 22, the lines split by naming (Figure 6.1). With aligned names they rise to 0.63–0.67 with demonstrations and 0.59 zero-shot; with flipped names they end at 0.45–0.52 with demonstrations and 0.45 zero-shot.

Finding 6.2. The critical layer is 23 of 28, and the names, not the demonstrations, drive it: zero-shot shows the same split.

The critical layer is 23 for random, `label_diversity`, `counter_prior` and SATA-MI, 24 for zero-shot and 25 for SATA. Zero-shot already has a final gap of 0.141; random demonstrations widen it to 0.192. The strategies built to work against the prior do not move the critical layer earlier; they only narrow the final gap (`counter_prior` 0.174, SATA-MI 0.119, SATA 0.104), which is the flattening seen in their behaviour (Figure 6.2).

Finding 6.3. The model's prior is linearly decodable from layer 2; the demonstrated rule hardly at all.

Probes read Qwen's zero-shot prediction from layer 2 onward with AUROC 0.80–0.87 under abstract names and 0.65–0.74 under named features (Figure 6.3). The rule label stays at 0.43–0.59, a little above chance only in the top layers under named features (0.59 at layer 28, aligned). A probe shows that information is present, not that it is used (Belinkov, 2022), and each task has its own rule, so a probe across tasks is a hard test; but the asymmetry is stark.

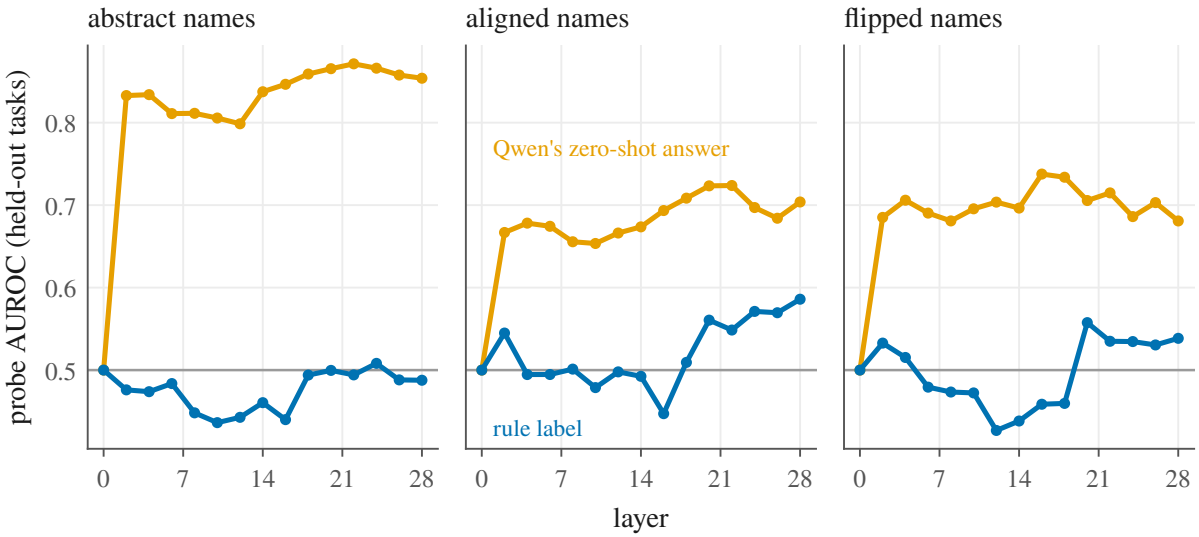


FIGURE 6.3. Linear probes on the residual stream (random demonstrations, tested on held-out tasks). Qwen's zero-shot answer is decodable from layer 2; the rule label stays near chance.

Finding 6.4. Within the steering we could test (one mean-difference direction, five layers, a 7B model with eight demonstrations), adding a "follow the demonstrations" direction moves the margins without making them more correct.

The steering vector is small, 1–2% of the residual norm, which is why the planned strengths ($\alpha \leq 4$) were extended to 32. For $\alpha \leq 16$, AUROC on the held-out tasks changes by at most ± 0.008 under both namings. At $\alpha = 32$ a few cells move by 0.02–0.04 in both directions, as the random direction does: damage from a large push, not a correction (Figure 6.4, top). Yet the margins move a lot, by 9.4 logits on average at layer 24 and $\alpha = 32$ (Figure 6.4, bottom). The reason is in how the vector was built. On conflict queries whose rule label is 0, following the demonstrations means a lower margin (0.56 against 6.08); on those whose rule label is 1, a higher one (5.25 against 0.41). Averaged over the two labels these halves point in opposite directions and mostly cancel, leaving a push of all margins together.

6.4 What this chapter does not show

The logit lens can be biased at early layers (Belrose et al., 2023); the probes avoid that bias and tell the same story, but neither is causal. Steering is the causal test, and it covers one family of directions only: a single mean difference, at five layers, applied to the query. Minder et al. (2025) find a context-versus-prior direction in Llama-3.1-8B; our result says that a direction built this way does not exist in a usable

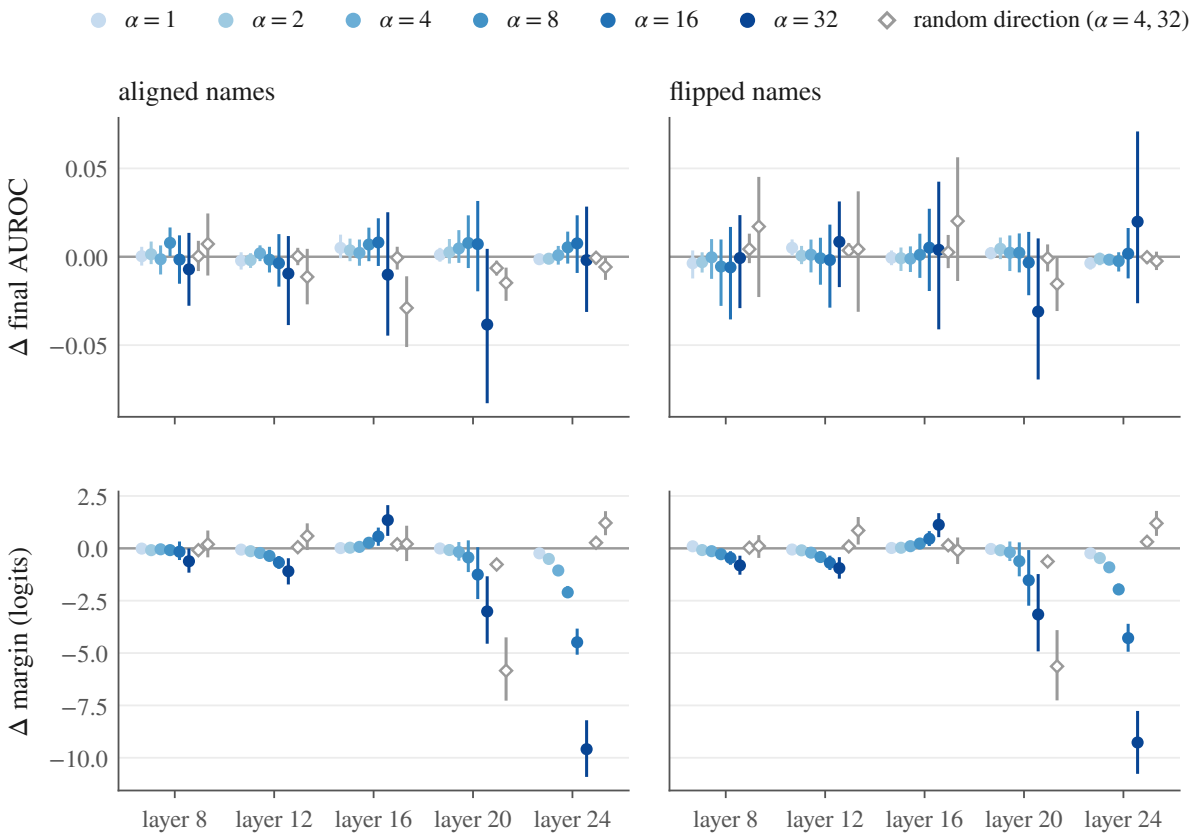


FIGURE 6.4. Steering on the 12 held-out tasks: change in final AUROC (top) and in the margin (bottom), with 95% intervals over tasks. Blue shades are the "follow the demonstrations" direction at increasing strength α; grey diamonds are a random direction of the same norm. Margins move by up to 9 logits while AUROC stays at zero: the direction shifts every answer together.

form for this task and model, not that none exists. A label-specific or learned direction, steering of the demonstration prefix, or a larger model could behave differently, and our budget did not reach them. The broader limitations of the study are the subject of section 7.4.

CHAPTER 7

Discussion

7.1 Answers to the research questions

RQ1: does in-context learning degrade under shift? Yes, and concept shift realised by naming degrades it most: flipped names cost 0.20 AUROC with random demonstrations (Chapter 4). Reversing the shortcut costs about half as much, and most of that cost is the model's numeric prior reading the shortcut in the query rather than anything the demonstrations taught. Covariate shift shows no cost in within-environment AUROC, but this is a property of the metric: the shift moves every score together, and a variance shift does hurt under flipped names.

RQ2: do different strategies help under different shifts? Only weakly. One interaction survives correction: neutralising the shortcut helps under its reversal more than under covariate shift. No strategy helps under flipped names, including the one designed for prior conflict.

RQ3: does the model rely on the right features, and why does the prior win? Under abstract names it relies on the shortcut; under aligned names, on the rule features; under flipped names, on the rule features in the names' direction. It cannot report which features it uses. The prior wins because it is present in the model's state from layer 2, while the demonstrated rule barely appears there, and the answer is formed in the last six layers, where the names decide it (Chapter 6).

RQ4: does a learned selector help, and do model-informed signals help it? Not on the instruction-tuned model. SATA and SATA-MI beat every strategy on stand-in learners and tie on Qwen2.5-7B-Instruct; neither the selection rule nor the prior signal changes this. SATA does help the base model, in an exploratory comparison.

7.2 The results in context

Wei et al. (2023) found that overriding semantic priors with in-context evidence emerges with scale. Our results are consistent with that picture and sharpen it for one model: for this pair of 7B models, the instruction-tuned one is held more tightly by its prior (the base model loses less to flipped names, and a selector helps it), although two models cannot show that instruction tuning is the cause, and the cost of conflict is set by the names more than by any choice of evidence. The graded picture of Krishna Kumar (2025) and Kossen et al. (2024), in which mid-sized models follow flipped labels part of the time, fits the fact that demonstration labels still matter here (gold minus shuffled labels: +0.06 to +0.11 AUROC) without being able to overturn the names.

Halawi et al. (2024) find that under false demonstrations, later layers switch to the demonstrated labels. We find a late critical layer as well, but the switch it marks is driven by the names, and zero-shot shows it too: the late layers are where the semantic prior is applied, not where the demonstrations take over. Minder et al. (2025) report a single context-versus-prior direction in Llama-3.1-8B; for our numeric task, a mean-difference direction of that kind was a bias rather than a switch. Nastl and Hardt (2024) found that restricting predictors to causal features did not improve generalisation on real tabular shifts; we find the same for demonstrations without the shortcut on ID queries, but a clear gain under reversal. Finally, learned selectors trained on model feedback (Rubin et al., 2022; Ye et al., 2023) are usually evaluated in distribution and on tasks the model already knows; our results suggest that a selector's value is bounded by how much the model reads the evidence at all.

7.3 Anticipated criticism

RQ1

"Covariate shift showing no effect means your covariate shift is too weak."

The shift moves two rule features by 1–2 standard deviations, and the model's scores move by 0.44–1.21 logits for the same prompts, so the shift reaches the model. What it does not do is reorder queries within an environment, which is all within-cell AUROC can see. We report this as a limitation of the metric (section 4.2), and a variance shift does show harm under flipped names.

"Naming is not concept shift; $P(y \mid x)$ never changes."

From the model's side it does: the same numbers under antonym names imply the opposite mapping, which is what concept shift means for a learner whose prior is semantic. We use the term in that operational sense and say so (section 3.3).

RQ2

"The strategies are weak straw men."

Each follows a published idea (Table 3.1), two use ground truth that a deployed selector would not have, and one is matched to the model's own prior. The comparator is balanced random, which controls the label balance that is known to bias predictions (Zhao et al., 2021). That most of them tie with it is the finding.

"24 tasks are too few to detect anything."

The intervals are about 0.1 AUROC wide for strategy contrasts, so effects below about 0.05 could be missed. The naming effect (0.20) and the interaction (0.11) were detected; for the ten selector contrasts, every upper bound is at most 0.09 AUROC, and eight are below 0.05. We do not claim that smaller effects are absent.

RQ3

"The logit lens is unreliable for early layers."

It can be biased there (Belrose et al., 2023). The final layer matches the model's output exactly, and the linear probes, which avoid that bias, tell the same story: the prior early, the rule weak throughout.

"Steering failing proves nothing about steering in general."

Agreed. We tested one family of directions (a label-averaged mean difference at five layers, applied to the query) on one 7B model with eight numeric demonstrations, within a small budget. We claim only that this direction acts as a bias at this scale, and we explain why it must: following the demonstrations means opposite things for the two labels.

"Self-reported importance is a known failure; why test it?"

Because it is the cheapest explanation a practitioner would ask for, and because the comparison with behavioural reliance under each naming shows that the self-report follows the names, not the model's behaviour.

RQ4

*"*SATA *failed because it was badly trained."*

It reached its own oracle's level on the stand-in learners and beat every strategy there; the base model benefits from the same checkpoints; and ranking instead of drawing does no better. The failure is in what the instruction-tuned model does with good evidence.

"The stand-in learners are poor proxies for the model, so the selectors were optimised for the wrong thing."

This is the main weakness of the selector chapters, and we measured it: the proxies rank strategies like Qwen at Spearman 0.68 (abstract) but only about 0.47 under named features, and the agreement check before the SATA-MI evaluation failed on exactly this. Training on the model's own scores, as datamodels would (Chang & Jia, 2023), was planned and not run within our budget.

"You chose the matched target and the drawing rule after seeing results."

Both were chosen on validation tasks and stand-in learners, before any selector result on the model; the original targets are reported as an ablation, and the drawing rule was checked on the model afterwards (section 5.3).

7.4 Limitations

Every negative result in this thesis holds for the conditions we could test, and those conditions were narrow.

- **Scale.** One model family at 7B: Qwen2.5-7B-Instruct and, exploratorily, its base model. Llama-3.1-8B-Instruct was dropped when it would not answer in format; a prefill would likely fix this, but was

not run. Larger models, which Wei et al. (2023) find follow contradicting evidence better, were out of reach.

- **Demonstrations.** $k = 8$, chosen on the pilot because $k = 32$ did not pass the learnability check. More demonstrations might give the rule more weight.
- **Tasks.** Synthetic numeric tasks with three load-bearing features, two domains and 24 evaluation tasks. Real tabular data, text features or other rule families may behave differently.
- **Metric.** Within-environment AUROC cannot see shifts that move every score together; balanced accuracy, its alternative, was distorted by calibration.
- **Proxies.** The selectors were trained against stand-in learners that track the model only moderately under named features.
- **Budget.** The later stages were sized to a few tens of US dollars of rented GPU time: SATA-MI's training was capped at 30 epochs, steering covered one direction family, and the datamodel variant of the selector (training on the model's own scores) and a reinforcement-learning selector were not run.
- **Decisions after data.** Some design decisions followed pilot or partial results: the random spurious direction, the switch to AUROC, the names chosen by measured contrast, the matched target. Each was made before the confirmatory family it affects was computed.
- **Numerics.** Rerunning identical prompts on a fresh GPU reproduced 407 of 432 units bit for bit; 25 differed by run-to-run GPU numerics, moving cell AUROC by 0.001 on average.

7.5 Future work

The most direct extension is scale: the same testbed and contrast families on larger models, where the prior's hold should be weaker, would show whether selection starts to matter as the model starts to read the evidence. Training the selector on the model's own scores of demonstration subsets (datamodels, Chang and Jia, 2023; Ilyas et al., 2022), possibly read at the critical layer, would remove the proxy problem. On the presentation side, a parked idea is to *turn* the prior rather than fight it: negate the features whose larger values go with label 0, in the demonstrations and the query alike, so that the "bigger means 1" habit points in the rule's direction. Finally, label-specific or learned steering directions, and steering of the demonstration prefix rather than the query, would test whether the rule can be made to reach the late layers at all.

CHAPTER 8

Conclusion

This thesis asked whether choosing the demonstrations can help a 7B language model follow them over its own priors under controlled distribution shift. To answer it we built a synthetic tabular testbed in which covariate shift, spurious reversal and concept shift can each be switched on alone, set the model's semantic prior against the rule by naming the features, and tested every claim against a contrast family frozen before its data.

The answer, for Qwen2.5-7B-Instruct with eight numeric demonstrations, is no. The feature names decide most of what the model predicts, more than any shift of the data. Hand-designed strategies help only where they neutralise a shortcut. A learned selector that beats every strategy on stand-in learners ties on the model, whether it reads only the data or also the model's measured prior, though it does help the base model, whose priors are weaker. Inside the model, the answer forms in its last few layers, the names decide it there, the prior is present from the start and the demonstrated rule hardly at all, and a single steering direction moves the answers without making them more correct.

Selection chooses evidence; it cannot make a model read evidence it largely ignores. Within the scale we could study, that is the binding constraint. The testbed, the pre-registered families and the diagnosis give future work, at larger scale and with the model's own scores as training signal, a controlled place to find out when it stops binding.

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APPENDIX A

Notation

Symbols are grouped by where they first appear. Every symbol is also defined in the text at its first use; this appendix is the single place to look one up. Three letters carry more than one meaning, always told apart by a subscript or superscript: $\sigma(\cdot)$ is the logistic function and σ_t a task's spurious direction; s is the shortcut's strength, s_j a load-bearing feature's direction in the rule, and s^{lex} a feature name's dictionary direction; w is the logit-lens label direction, and $\hat{w}$ the reader's fitted weights.

Symbol	Meaning
Tasks and data	
$f_0, \ldots, f_9$	The ten features of a task, in the generator's numbering; also their abstract display names
$x \in \mathbb{R}^{10}$, x_j	A row's feature vector and its j-th feature, standardised with the pool's statistics
C, $\lvert C\rvert = 3$	The load-bearing (causal) features: the label depends on these and nothing else
x_C	The load-bearing part of x
$c_j = u_j r_j$	Linear coefficient of load-bearing feature j, with $u_j \sim \mathrm{U}(1, 2)$ and a random sign $r_j \in \{+1, -1\}$
$s_j \in \{+1, -1\}$	Direction of load-bearing feature j in the rule ($s_j = \operatorname{sign} c_j$ for linear tasks)
$\sigma(z) = 1/(1 + e^{-z})$	The logistic function
Maj	Majority vote over three signs (tree rules)
y_{clean}, y	A row's rule label before and after label noise
$\eta = 0.02$	Label-noise rate: the probability that $y \neq y_{\mathrm{clean}}$
$s \sim \mathrm{U}(0.80, 0.90)$	Spurious strength: the probability that the shortcut's sign agrees with the label
$d = \Phi^{-1}(s)$	The shortcut's class-conditional mean, with Φ the standard normal distribution function
$\sigma_t \in \{+1, -1\}$	Task t's spurious direction: whether a large shortcut value indicates label 1 or label 0

Symbol	Meaning
ε	Standard normal noise in the shortcut
π_t	Task t's permutation of feature positions (the displayed order)
T, n	Number of tasks; queries per task and environment ($T = 24$, $n = 20$ in evaluation)
δ	Size of a shift or a nudge, in standard deviations
Naming, prompts and the model	
N	A naming: abstract, aligned or flipped
$s^{\text{lex}} \in \{+1, 0, -1\}$	A display name's dictionary direction ($+1$ for the first member of an antonym pair, -1 for the second, 0 for neutral and abstract names)
k	Demonstrations per prompt ($k = 8$)
ℓ_0, ℓ_1	The model's logits for the label tokens `0` and `1`
$m = \ell_1 - \ell_0$	The label margin, equal to $\log p(\texttt{1}) - \log p(\texttt{0})$
$m_0(x; N)$	The model's zero-shot margin on row x under naming N: its prior
$\tilde{m}_0(x_i; N)$	The measured prior of pool row i, centred on the pool median
C_t	Task t's prior–data conflict under naming N: the share of pool rows whose label disagrees with the prediction of the centred prior, $\mathbf{1}[\tilde{m}_0 > 0]$
Evaluation and statistics	
AUROC	Area under the ROC curve of the margin against the label, within one cell (task, strategy, seed, naming, environment)
$B = 2000$	Bootstrap draws
C1a–C3b	The confirmatory contrasts for **RQ1**–**RQ3** (Table B.1)
R1–R6, R7–R10	The confirmatory contrasts for SATA and SATA-MI (Table B.1)
π_{true}, π_{behav}, π_{self}	Feature importance: ground truth, behavioural (margin drop when the feature is replaced), and the model's self-report
$\rho(\cdot, \cdot)$	Spearman rank correlation
g_j	Directional sensitivity: $m(x + \delta e_j) - m(x - \delta e_j)$ with $\delta = 0.5$
DFI	Demo-following index: the share of (query, load-bearing feature) pairs with $\operatorname{sign} g_j = s_j$
Selectors	
p_θ	The selector's softmax distribution over pool rows, with parameters θ
q	Its training target: a distribution over pool rows
$\mathrm{KL}(q \,\|\, p_\theta)$	The training loss
λ	Weight of the counter-prior bonus in SATA-MI's targets ($\lambda = 0$ was kept)

Symbol	Meaning
$\hat{w}$	The reader's ridge-logistic weights, fitted on the eight chosen demonstrations
a_0, a_1, a_2	The reader's coefficients: $P(\text{Qwen says } 1) = \sigma(a_1 \hat{w}^\top x + a_2 m_0 + a_0)$
Inside the model	
$L = 28$, l	Number of decoder layers; a layer index ($l = 0$ is the embeddings)
h_l	The last-position residual state after layer l
W_U	The unembedding (output) matrix
$w = W_U[1] - W_U[0]$	The label direction: the difference of the two label rows
$\mathrm{RMSNorm}$	The model's final normalisation layer
m_l	The logit-lens margin at layer l: $w^\top \mathrm{RMSNorm}(h_l)$ for $l < L$, and $m_L = m$
AUROC_l	AUROC of m_l against the rule label
v_l	The steering vector at layer l
α	Steering strength: $h_l \leftarrow h_l + \alpha v_l$

APPENDIX B

Design details

This appendix gives the definitions and properties that the chapters rely on.

B.1 Tasks and environments

Rows. Features are drawn as $x \sim \mathcal{N}(0, I_{10})$, the environment's perturbation is applied, the clean label is computed from the load-bearing features C, the shortcut x_8 is generated from the clean label, $x_9 \sim \mathcal{N}(0,1)$, and labels are flipped with probability $\eta = 0.02$. Features are standardised with the pool's statistics and shown in a random order π_t per task, so no role can be read from a position.

Rules. A *linear* task has $y_{\text{clean}} \sim \text{Bernoulli}\big(\sigma(c^\top x_C)\big)$ with $c_j = u_j r_j$, $u_j \sim \text{U}(1,2)$, $r_j \in \{\pm 1\}$; $P(y_{\text{clean}} = 1 \mid x)$ is strictly monotone in each x_j, so feature j has the direction $s_j = \operatorname{sign} c_j$. A *tree* task splits each of three features at zero and labels the eight leaves; the leaf tables allowed (complement-antisymmetric, neither a dictator nor parity) are exactly the signed majorities $y_{\text{clean}} = \mathbb{1}\big[\text{Maj}(s_1 x_1, s_2 x_2, s_3 x_3) > 0\big]$, so tree tasks are monotone too.

The shortcut. With $d = \Phi^{-1}(s)$, $x_8 = \sigma_t\big((2y_{\text{clean}} - 1)d + \varepsilon\big)$, $\varepsilon \sim \mathcal{N}(0,1)$.

PROPOSITION B.1 (Sign agreement). $P\big(\operatorname{sign}(\sigma_t x_8) = \operatorname{sign}(2y_{\text{clean}} - 1)\big) = s$*; under spurious reversal* ($d \to -d$) *it is* $1 - s$.

PROOF. For $y = 1$, $P(\sigma_t x_8 > 0) = P(Z > -d) = \Phi(d) = s$; $y = 0$ is symmetric. Reversal replaces d by $\Phi^{-1}(1-s) = -d$. □

Covariate shift. Two load-bearing features i, j are moved in opposite label directions and a distractor $d \notin C$ at random: $\delta_i = s_i m/|c_i|$, $\delta_j = -s_j m/|c_j|$, $\delta_d \sim \pm\text{U}(1,2)$, with the push m chosen so that every shifted feature moves by 1–2 standard deviations (for trees $|c| = 1$, $m \sim \text{U}(1,2)$).

LEMMA B.2 (The label rate is unchanged). *(a) For linear tasks, $c^\top(x+\delta) = c^\top x$ for every row. (b) For tree tasks, $P(y_{\text{clean}} = 1)$ stays $1/2$.*

PROOF. (a) $c_i\delta_i = m$, $c_j\delta_j = -m$, and the distractor carries no weight. (b) The three votes are independent. After the shift the two moved votes are 1 with probabilities $p_i = \Phi(m)$ and $p_j = 1-\Phi(m)$, and the third with $\frac{1}{2}$, so the majority is 1 with probability $p_ip_j + \frac{1}{2}[p_i(1-p_j)+(1-p_i)p_j] = \frac{1}{2}(p_i+p_j) = \frac{1}{2}$. □

B.2 Naming

Each domain has antonym pairs (p, q), where p is associated in everyday language with label 1 and q with label 0, and a set of weak-prior names. Each load-bearing feature j is assigned a pair: *aligned* names it p if $s_j = +1$ and q if $s_j = -1$; *flipped* uses the other member. Every other feature gets a weak-prior name, the same in both conditions.

PROPOSITION B.3 (Identification). *For a fixed task, demonstration set and query, the aligned and flipped prompts differ only in the names of the three load-bearing features. A predictor that ignores names has the same expected accuracy under both, so a difference in the model's accuracy is attributable to the meaning of the names.*

B.3 Counter-spurious selection

Balancing the four (label, shortcut agrees) cells is not enough: agreeing rows lie further from zero ($\mathbb{E}[|x_8| \mid \text{agree}] = d + \phi(d)/\Phi(d) > \mathbb{E}[|x_8| \mid \text{disagree}] = \phi(d)/(1-\Phi(d)) - d$), so x_8 still correlates with the label (about 0.33 at $s = 0.85$). `counter_spurious` therefore takes, within each label, $k/4$ clean disagreeing rows and pairs each with the clean agreeing row of the same label nearest in $|x_8|$ (Rubin, 1973; Stuart, 2010). With perfect matching, sign agreement is $\frac{1}{2}$ and $\mathrm{Cov}(x_8, 2y-1) = 0$; on the evaluation tasks the implementation gives $\mathrm{corr}(x_8, y) = 0.005$. SATA's matched target (section 5.2) is each row's probability π_i of appearing in such a set built for the query (the query's tree regime first), normalised: $q_i = \pi_i / \sum_j \pi_j$; label-noise rows have $\pi_i = 0$, and nothing reads the query's label. SATA is trained with the loss $\mathrm{KL}(q \,\|\, p_\theta)$.

B.4 Scoring and caching

LEMMA B.4 (Margin). *With $p_1 = e^{\ell_1}/(e^{\ell_0} + e^{\ell_1})$ and $m = \ell_1 - \ell_0$, $p_1 = \sigma(m)$, and m is the same whether computed from logits or from full-vocabulary log-probabilities.*

PROOF. $p_1/(1 - p_1) = e^{\ell_1 - \ell_0}$; log-probabilities subtract a common log-sum-exp, which cancels. □

PROPOSITION B.5 (Prefix caching is exact). *For a decoder-only transformer with causal attention, running the prefix P once and each suffix S against its cached keys and values gives the same final-position logits as a full pass on $P\|S$, provided* $\mathrm{tok}(P\|S) = \mathrm{tok}(P)\|\mathrm{tok}(S)$.

PROOF. States at positions $< |P|$ depend only on those positions' tokens (attention reads positions $\leq t$), so appending S does not change them, and the states of S are computed from the same inputs either way. The runner checks the tokenisation condition for every suffix. □

LEMMA B.6 (Logit lens at the last layer). *With $w = W_U[1] - W_U[0]$ and h_L the final (normalised) state, $m_L = w^\top h_L = \ell_1 - \ell_0 = m$.*

B.5 Statistics

Cells and contrasts. AUROC is computed within each cell (task, strategy, seed, naming, environment) over its 20 label-balanced queries and averaged over seeds. A contrast is a per-task difference of such values (for example, strategy A minus strategy B on the same queries, or ID minus a shifted environment), averaged over the 24 tasks.

Hierarchical bootstrap. Each of $B = 2000$ draws resamples tasks with replacement and, within each resampled task, its positive and negative queries separately; the percentile interval of the resampled estimates is reported, with one-sided p-values from the share of draws on the wrong side of zero. Because tasks differ in their rules, the variance of the mean is dominated by the between-task term ($\sigma_T^2/T + \sigma_W^2/(Tn)$; Cochran, 1977), so tasks are the unit of resampling.

Multiplicity. Holm's step-down correction (Holm, 1979) is applied within each pre-registered family: ten contrasts for **RQ1–RQ3**, six for SATA (R1–R6) and four for SATA-MI (R7–R10). Table B.1 lists every confirmatory estimate.

TABLE B.1. Every confirmatory contrast (Qwen2.5-7B-Instruct, 24 tasks; one-sided hierarchical bootstrap, Holm within each family). C1–C2 and R are differences in AUROC; C3a in the demo-following index; C3b in Spearman ρ. Bold: significant after correction.

	Contrast	Estimate [95% CI]	Holm p	Tasks
C1a	ID − covariate (random, abstract)	$+0.024$ [-0.044, 0.093]	0.95	15/24
C1b	ID − spurious reversal (random, abstract)	$+0.115$ [0.017, 0.218]	0.063	16/24
C1c	**aligned − flipped (random, ID)**	$\mathbf{+0.204}$ [0.131, 0.280]	**0.005**	22/24
C2a	counter_spurious − label_diversity, reversal	$+0.035$ [-0.019, 0.086]	0.59	16/24
C2b	similarity − label_diversity, covariate	-0.029 [-0.110, 0.048]	1.00	11/24
C2c	feature_range − label_diversity, covariate	-0.078 [-0.141, -0.018]	1.00	6/24
C2d	counter_prior − label_diversity, flipped, ID	$+0.023$ [-0.015, 0.064]	0.70	7/24
C2e	**C2a under reversal − under covariate**	$\mathbf{+0.111}$ [0.033, 0.196]	**0.024**	17/24
C3a	**demo-following index, abstract − flipped**	$\mathbf{+0.123}$ [0.079, 0.175]	**0.005**	23/24
C3b	$\rho(\pi_{\text{true}}, \pi_{\text{behav}})$, abstract − flipped	$+0.026$ [-0.141, 0.174]	1.00	13/24
R1	sata − label_diversity, reversal	$+0.032$ [-0.024, 0.089]	0.79	14/24
R2	sata − label_diversity, covariate	-0.096 [-0.154, -0.041]	1.00	5/24
R3	sata − label_diversity, ID	-0.040 [-0.102, 0.023]	1.00	8/24
R4	sata − similarity, mean of 3 environments	-0.020 [-0.064, 0.026]	1.00	11/24
R5	sata − sata without the query, mean of 3	-0.025 [-0.063, 0.010]	1.00	10/24
R6	sata − label_diversity, flipped, ID	-0.014 [-0.077, 0.048]	1.00	13/24
R7	sata_mi − sata, flipped, ID	$+0.005$ [-0.050, 0.062]	1.00	11/24
R8	sata_mi − sata_names, flipped, ID	-0.010 [-0.066, 0.047]	1.00	10/24
R9	sata_mi − counter_prior, flipped, ID	-0.032 [-0.099, 0.036]	1.00	11/24
R10	sata_mi − sata, flipped, mean of 3	$+0.001$ [-0.031, 0.032]	1.00	12/24

B.6 Reproducibility

The code, configuration and pre-registered contrast families accompany this thesis, with tests covering the invariants of the generator, the selection engine, the model runner and the probes. Every prediction is stored with its demonstration rows, order and prompt, so every number and figure here can be recomputed from the result files.[5]

[5] **TODO** *Add the repository link.*